

RYO-CHAN Hackathon 2026 — Project Submission Form

Filled against the organiser's own template, downloaded from `https://ryobuild.com/RYOCHAN-Hackthon-Project-Submission-Form.pdf` on 2026-09-07 and committed unchanged beside this file as `RYOCHAN-Hackathon-Project-Submission-Form-BLANK.pdf`. That PDF has no interactive fields, so this is the same form filled in, in the same order, and rendered to `project-submission-form.pdf` by `scripts/submission_pdf.py`.

Members

Total Members		1
Name	Role	Email
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Overview

Field	Value
Project	Nota
Track	Track 1 (Autonomous Agents), Track 2 (Dashboards & Interfaces) and Track 3 (New Skills). Each is judged independently.
Problem	An AI opinion about a market is unfalsifiable. You cannot tell whether the number in a sentence came from the evidence or from the model, whether a missing feed was quietly treated as zero, or whether the answer would be the same tomorrow. Nothing in a chat log can be re-run, so nothing in it can be audited, and a confident paragraph is indistinguishable from a careful one.
Solution	Nota turns each decision into a receipt that reproduces. Three specialised agents argue over live RYO evidence and cite dotted paths into it; a judge weighs them by each agent's own Brier score from past calls; risk sizing is pure ATR arithmetic with no model in it. <code>nota replay <id></code> rebuilds the decision from the stored evidence and the model output cached beside it, keyed by the model that produced them, and prints <code>identical: True</code> . That runs with no API key and no network, so anyone who clones the repository can check the claim. Nota also audits RYO itself: it recomputes RYO's RSI and ATR from public candles with Wilder's method, checks RYO's price against three exchanges, and compares its own sizing against RYO's published trade plan. When the sources disagree, the judge refuses to size a trade at all. And it keeps a record RYO does not: the RYO Verdict Scorecard locks every <code>deep_analysis</code> verdict and trade plan for 25 majors once a day and settles each plan on OKX's hourly candles (stop or +1R first, within 24 h and 72 h), so a reader can see whether RYO's verdict changes how often RYO's own plan works.
Key Features	Council of three agents with dotted-path citations, and citations pointing at absent evidence are dropped in code rather than trusted. Judge weighted by historical Brier score. Pure-ATR sizing that blocks rather than guesses. Replayable receipts, verifiable from the browser or the CLI. A derivatives gate: <code>positioning_check</code> rules each field of RYO's derivatives block citable or not without guessing RYO's undocumented definitions (the same value on two other tokens that day is <code>not_token_specific</code> ; on 18 September one 24 h open-interest change repeated across ETH, SOL, WIF and ONDO), and a withheld field is masked in the prompt and cannot be cited. The RYO Verdict Scorecard at <code>/scorecard</code> : daily locks at least 20 h apart, independent settlement on OKX 1-minute and hourly candles, day-matched Mantel-Haenszel comparisons of verdicts, confluence states and RYO's momentum gate on both the target-first rate and the return at the horizon, with

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	a within-cluster permutation p-value and no call before 10 shared days, a per-lane record of how often RYO's answer came back available, and a coverage count where every failed lock is a line; on day 0, 7 of 25 plans were long under a <code>cautious</code> verdict. Every agent is scored against the token's own base rate (how often it closed higher seven days later on same-volatility days, counted before the decision), not only against a coin flip. Diff-first dashboard that ranks what changed against the previous receipt by impact. Nine research skills in RYO's own envelope (the seven above plus <code>liquidity_check</code> on DefiLlama stablecoin supply and chain TVL, and <code>crowd_odds</code>, the Polymarket and Kalshi baseline the judge is scored against), served both on RYO's <code>/api/skills</code> paths and as a real MCP server at <code>POST /mcp</code> with tools and resources. <code>nota watch</code> autonomous loop with Telegram and Discord publishing. Public backing on receipts, scored when the call resolves. An A2A 1.0 Agent Card at <code>/well-known/agent-card.json</code> and a <code>SendMessage</code> endpoint at <code>/a2a</code> over the same skills. Open data: Atom and JSON feeds of receipts, outcomes and settled plans, CSV exports of the scorecard and of every resolved decision, a schema.org Dataset on <code>/scorecard</code>, a sitemap and robots.txt. Open Graph receipt cards, markdown and JSON exports, <code>llms.txt</code>, a captioned walkthrough, keyboard-first accessible interface in light and dark. The landing page states no number of its own: it draws its marks, its verify target and its failure panel from <code>/api/decisions</code>, and a test asserts there is no evidence hash written into the page at all. That failure panel picks, from the newest 50 receipts, the one where the most RYO sections failed (the newest on a tie), and shows it with that receipt's own availability rows and warnings, so it always shows the worst failure on record and how the independent skills kept answering through it. Every snapshot the ledger cycle pushes is attested with Sigstore (GitHub artifact attestations) and every lock and receipt is anchored in Bitcoin through OpenTimestamps.
Target Users	Traders and analysts who need to justify a call rather than only make one; RYO itself, which can mount the nine skills as routes or call them over MCP; and hackathon judges, who can verify every claim on this page from outside the project.
Scope	Read-only research over RYO's six builder tools plus independent public sources. Practice trades only: sizes, stops and targets are recorded and marked against later prices, and no order is ever placed, signed or routed. Ships with a ledger snapshot that starts at four receipts made on live RYO evidence on 2026-09-07 and grows: a GitHub Actions cycle scores every decision that reached its seven-day horizon within the hour and commits the snapshot back, which is what turns the judge's Brier weighting from a design into a measurement. The same cycle locks the scorecard's 25 plans twice a day (the second pass only retries what failed) and settles the ones whose window has closed every hour; that needs the builder key for the lock and nothing for the settlement. Scoring needs no key of any kind - the price comes from the OKX close at the horizon, or the exchange median labelled late - so the ledger kept calibrating while the builder key answered 401 (10 to 18 September), and the cycle now ends on <code>nota health --strict</code>, so a dead or expiring key turns the run red instead of leaving it green with no locks.
Limitations	Stated plainly rather than hidden. A council decision needs one LLM key; live RYO evidence needs the builder key; verification and the skills need neither. X's public syndication endpoint throttles hard and has tightened: it hit data-centre addresses first, and as of 2026-09-10 an ordinary connection gets <code>syndication HTTP 429</code> as well, so <code>x:</code> voices are best effort while <code>tg:</code> and <code>bs:</code> answer from both. <code>TAVILY_API_KEY</code> turns that into a search-backed fallback the envelope labels as undated; unset, the warning names the missing key rather than the voice going quietly silent. Brier weights only move once calls reach their seven-day horizon, and then only by $n/(n+20)$ of the way, because five scored calls (all Nota had by 18 September) are far too few to trust; the dashboard counts how many are independent weeks and says so beside every such table. The hosted deployment reads its ledger from a snapshot because a serverless filesystem cannot be written, so it is only as fresh as the last cycle's commit (at most an hour for settlements); backings are kept in Postgres (Neon, free tier) so the public backing feature works there, while every other write still belongs to the CLI. With no <code>DATABASE_URL</code> at all — a plain clone — backings go to the ledger's own table, so nothing needs a database to run or to test. The receipt diff ranks changes with a documented heuristic, not a model. The scorecard began on 18 September, so at judging it holds about two weeks of lock days in what has so far been one market regime; its comparison

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	between confluence states will most likely still read "not yet distinguishable", and the page says so instead of drawing an interval from too few days. Settlement splits an hour that touched both levels on its 1-minute candles; a stop and a target inside the same minute stay <code>ambiguous</code> rather than ordered by guess, and the bracket is re-anchored to OKX's price because RYO and OKX are two feeds (a lock more than 2% apart is kept and never settled).

Tech Stack

Field	Value
Frontend Stack	No framework and no build step: four hand-written pages (the landing in English and Japanese, the receipt dashboard, the scorecard, and the walkthrough) of semantic HTML with native CSS and vanilla JavaScript, served by the same FastAPI process. One palette expressed with CSS <code>light-dark()</code> so light, dark and system all work from one definition, <code>prefers-reduced-motion</code> honoured, keyboard navigation and visible focus throughout. The hero mark is an inline SVG generated from the receipt's own evidence hash, and <code>/demo</code> pairs the walkthrough with a clickable transcript built from the recorder's own timings.
Backend Stack	Python 3.12, FastAPI and uvicorn, Pydantic v2 for every boundary type, httpx for outbound calls, SQLite in WAL mode as the ledger (evidence packs, LLM output cache, decisions, outcomes, and backings when no Postgres is configured), psycpg 3 against Neon Postgres for backings on the hosted deployment, Typer for the <code>nota</code> CLI, Pillow for receipt cards, defusedxml for RSS, vaderSentiment for lexicon scoring. Deployed on Vercel as a framework-detected FastAPI app.
AI Model(s)	Provider-agnostic by design. Two adapters: Anthropic, and any OpenAI-compatible endpoint. Every shipped receipt was produced through Venice with <code>qwen3-235b-a22b-instruct-2507</code> . Replay is keyed by the model that produced a receipt rather than by whatever is configured now, which is what let an earlier receipt made under <code>anthropic/claude-haiku-4.5</code> keep verifying after the provider changed. Model outputs are cached by <code>(evidence hash, role, prompt version, model)</code> .
Other Technologies	Model Context Protocol in both directions: Nota is an MCP client of RYO (JSON-RPC <code>tools/call</code> , selectable with <code>RYO_TRANSPORT=mcp</code>) and an MCP server of its own over Streamable HTTP at <code>POST /mcp</code> , negotiating protocol versions 2026-07-28 through 2024-11-05 and serving both <code>tools/*</code> and <code>resources/*</code> . Public data sources, all keyless: OKX public market and derivatives data (daily and hourly candles, perp premium, open interest in coins, long/short account ratio), Hyperliquid's public info endpoint, CoinGecko, Coinbase, Kraken, alternative.me Fear & Greed, t.me/s channel previews, the Bluesky public AppView, X's public syndication endpoint, and CoinDesk, Cointelegraph, The Block and Decrypt RSS. Tavily or Venice web search when a key is present. Playwright drives the browser QA suite and records the walkthrough, <code>edge-tts</code> (GPL-3.0) reads the narration with a Microsoft neural voice and no key, and Remotion (Remotion License, free at this team size) composes the two; none of the three ships any code into the application.

Repository / Demo

Field	Value
Github Repository	https://github.com/RYO-Digital/ryochan-hackathon_repository-235 (the repository issued for this project; all final code is on <code>main</code>). Public mirror of the same history: https://github.com/PugarHuda/nota — listed in the official MCP registry as <code>io.github.PugarHuda/nota</code> , published 2026-09-10 from the repository over GitHub OIDC, so any MCP client can discover this server without being told its URL: https://registry.modelcontextprotocol.io/v0.1/servers?search=io.github.PugarHuda/nota

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Demo Video	https://nota-ryo.vercel.app/demo — a narrated 3:36 walkthrough (re-recorded 2026-10-01, after ten days of settlements) with a clickable transcript, served by the project itself, public and with no password; the bare file is at https://nota-ryo.vercel.app/demo.mp4 and is committed at <code>nota/static/demo.mp4</code> . It was recorded against the running app: a cursor travels to whatever the voice is describing and a box frames it, the verify button is pressed on camera, and every figure spoken is one this deployment produces on demand.
Documentation	<code>README.md</code> in the repository is the primary document: how a decision is made, the honesty rules the code enforces, the MCP server, the dashboard, the skills, and how to evaluate everything with zero keys. <code>docs/skills/SKILL-SPEC.md</code> is the skill contract; <code>docs/HACKATHON-ANALYSIS.md</code> records what was verified about the platform; <code>docs/SUBMISSION.md</code> tracks this submission. Live: https://nota-ryo.vercel.app and https://nota-ryo.vercel.app/lms.txt

Testing Information

Field	Value
How to run the Project	Fastest: https://nota-ryo.vercel.app/judges maps each track to its rubric with links to the exact receipts, and the hosted dashboard's "Run the council live" convenes the council on fresh RYO evidence (capped per visitor). Locally: Clone the repository, <code>uv sync</code> , then <code>NOTA_DB=data/demo.db uv run nota serve</code> and open http://127.0.0.1:8000 for the overview or http://127.0.0.1:8000/app for the dashboard itself. That serves the shipped ledger of receipts made on live RYO evidence, so the dashboard, the replay verification, the receipt cards, the exports and the skill runner all work before any key is set.
Prerequisites	Python 3.12 or newer and uv . No database server, no Docker, and no Node to run or evaluate the project. For the browser tests only: <code>uv run playwright install chromium</code> . Node is needed only to re-render the walkthrough video, which ships built.
Installation Steps	1. <code>git clone <repository> && cd nota</code> 2. <code>uv sync</code> 3. <code>cp .env.example .env</code> 4. optionally fill in the keys below 5. <code>uv run nota health</code> to confirm RYO is reachable.
Environment Variables	All optional for evaluation; <code>.env.example</code> lists every one with comments and no values. <code>RYO_MCP_KEY</code> for live RYO evidence, with <code>RYO_MCP_URL</code> and <code>RYO_TRANSPORT</code> (<code>rest</code> or <code>mcp</code>). One LLM key for the council: <code>ANTHROPIC_API_KEY</code> , or <code>NOTA_LLM=openai</code> with <code>OPENAI_BASE_URL</code> and <code>OPENAI_API_KEY</code> for any OpenAI-compatible provider, plus <code>NOTA_MODEL</code> and <code>NOTA_MAX_TOKENS</code> . <code>ANTHROPIC_AUTH_TOKEN</code> stands in for the Anthropic key and <code>VENICE_API_KEY</code> for the OpenAI-compatible one. <code>NOTA_DB</code> selects the ledger, <code>NOTA_READONLY=1</code> opens it immutably, <code>DATABASE_URL</code> sends backings, handle claims and the shared rate limit to Postgres, <code>NOTA_PUBLIC_URL</code> sets the base for permalinks, canonical links, feeds and the A2A card. Optional: <code>NOTA_VOICES</code> , <code>TAVILY_API_KEY</code> , <code>COINGECKO_API_KEY</code> , <code>TELEGRAM_BOT_TOKEN</code> , <code>TELEGRAM_CHAT_ID</code> , <code>DISCORD_WEBHOOK_URL</code> . A test fails if the code reads a variable <code>.env.example</code> does not list. No real key is committed anywhere, and the git history was scanned to prove it.
Build Command	None. There is no build step: no bundler, no transpiler, no generated assets. <code>uv sync</code> installs the locked dependencies from <code>uv.lock</code> and the app runs from source.
Run Command	<code>uv run nota serve</code> for the interface and the API, or <code>uv run nota decide SOL</code> for one decision, <code>uv run nota watch SOL,BTC --every 3600 --notify</code> for the autonomous loop, <code>uv run nota replay <id></code> to verify a receipt, <code>uv run nota skill run price_crosscheck '{"symbol":"SOL"}'</code> for a skill. <code>uv run nota lock</code> and <code>uv run nota settle</code> for the scorecard. <code>uv run nota --help</code> lists every command, and the <code>skill</code> group holds <code>skill spec</code> and <code>skill run</code> .

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Test Command	<code>uv run pytest -q</code> . GitHub Actions runs it on every push and pull request, browser tests included, and production is deployed only after it passes on <code>main</code> . It needs no network and no keys: unit tests, API tests, MCP server and transport tests, A2A agent tests, feed, sitemap and CSV tests, an axe-core accessibility gate over every page in both themes at desk and phone widths, and browser checks that drive a real uvicorn with Playwright over the shipped ledger, failing on any console error or failed request — including one that plays the shipped demo video and refuses a transcript chapter that starts past its end. Without Chromium installed the browser checks skip with a clear reason rather than failing.
Test Account(s)	None needed. Nothing in Nota has a login: there are no accounts, no sessions and no wallets. Backing a receipt asks only for a handle. On the hosted deployment backings are live, stored in Neon Postgres: the first backing under a handle claims it and returns an edit token once (the browser keeps it), and a later backing under that handle without the token is refused with a 409.

Declarations

- **No secret has ever been committed**, and the organiser's review of git history was anticipated. Re-checked on 2026-09-20 across every commit in the repository: the exact values of every key in the local `.env` and `.env.local` (builder key, LLM key, Neon connection strings and password, Vercel token) were searched for in every patch and found in none. The first scan, at 49 commits and 448 blobs, ran against twelve credential patterns (RYO builder keys, Anthropic, OpenAI, OpenRouter, Venice, Tavily, Telegram bot tokens, Discord webhooks, AWS, GitHub and Vercel tokens, PEM private keys). The single match is `ryo_mcp_your_private_key`, the placeholder inside RYO's own builder guide that this repository quotes at `docs/MCP-Builder-Guide.md`. No `.env` or credential file was added in any commit; `.env.example` carries names only.
- **No fabricated data is presented as real**. Every receipt prints its own source label. `live` means RYO answered; `recorded` means a real RYO response captured with `nota record` and replayed. There is no third kind: the hand-built fixtures are gone, and the test fixtures are real RYO answers and refusals too (the one veto case is a real answer with only its veto fields edited, and its `_note` says so). `data_mode` is carried through from RYO per section, a value that could not be fetched stays `null` and is never turned into zero, and the risk layer refuses to size a trade on evidence RYO marks `simulated`.
- **Third-party code and resources are disclosed** in the README. No starter template was used; the repository began empty and every line of application code was written during the hackathon. Libraries: `httpx`, `pydantic`, `fastapi`, `uvicorn`, `typer`, `python-dotenv`, `anthropic`, `vaderSentiment`, `defusedxml`, `pillow`; development only: `pytest`, `respx`, `playwright`, `axe-playwright-python`. The `taste-skill` design ruleset was read while reshaping the interface and is gitignored rather than vendored, so no file of it ships here.
- **Read-only**. The RYO surface used is read-only, and Nota never places, signs or routes an order. Practice positions exist only in the local ledger and are labelled as practice everywhere they appear.